**EXAMPLE : 2.24**

Consider the following statements for plasma arc welding:

1. In tranferred arc method the workpiece is connected to torch electrically whereas in non-transferred arc the workpiece is not electrically connected to torch.
2. In transferred arc method the workpiece is not connected to torch electrically.
3. The transferred arc is usually employed for welding or cutting.

Which of the above statements is/are incorrect?

(a) 1 and 2 (b) 2 only

(c) 3 only (d) 1 only

**Solution: (b)**

Statement 1 and 3 are correct.

## 2.21 DEFECTS IN WELDING

The ideal weld should be such that adequate fusion exists between the filler metal and edge preparation together with good penetration.

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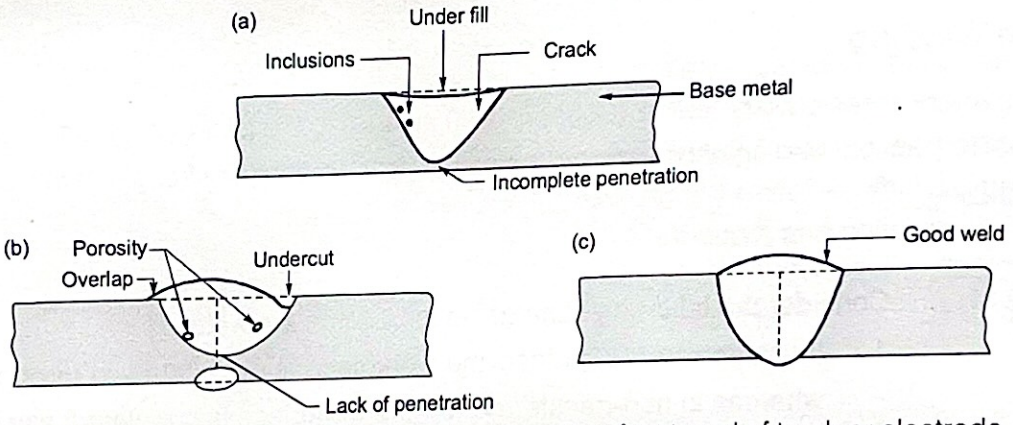
### 2.21.1 Porosity

- Porosity is caused by the entrapment of gas bubbles by the freezing dendrites during the cooling of the weld pool and occur essentially only in the weld metal.
- These are generally in the shape of spheres or elongated pockets.
- This can be reduced by proper selection of filler metals, by preheating the weld area, by proper cleaning of the weld zone, by providing shielding gas properly.
- The distribution of porosity in the weld zone may be random or the porosity may be concentrated in a certain regions.

### 2.21.2 Slag inclusion

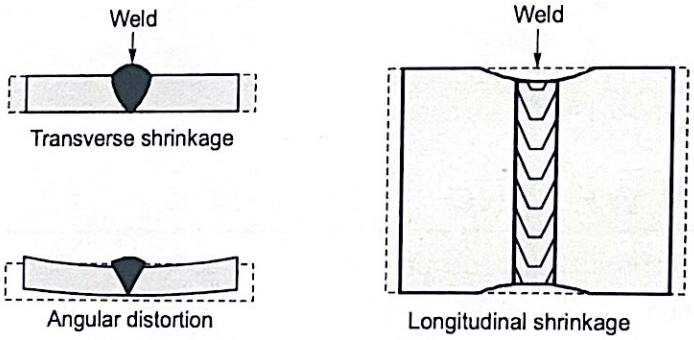
- Inclusions may be caused by compounds such as oxides, fluxes and electrode coating material, which are trapped in the weld zone.
- This can be reduced by cleaning the weld bead, by providing enough shielding gas.
- Slag inclusion can be prevented by
  (a) Cleaning the weld-bead surface before the next layer is deposited, by means of a wire brush in multipass welding,
  (b) Providing enough shielding gas,
  (c) Providing sufficient heat input

### 2.21.3 Incomplete fusion and penetration




- It is usually caused by insufficient heat input and too fast travel of torch or electrode.
- Incomplete penetration occurs when the depth of the weld joint is insufficient.
- This can be avoided by raising the temperature of the base metal.
- This can also be reduced by increasing the heat input, reducing speed, changing joint design and ensuring proper fit.

### 2.21.4 Under Cut




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- It is the melting or burning away of the base metal at the toe of the weld as sharpness or notch.
- Undercut can act as stress raiser and can reduce the fatigue strength of the joint.
- In such cases it may lead to premature failure.
- These undercuts are generally due to excessive welding speeds.

### 2.21.5 Overlapping

It is the protrusion of weld metal beyond the root of the weld and occurs due to insufficient travel speed which permits the weld puddle to move ahead of electrode, also due to insufficient angle.

### 2.21.6 Cracks

- Cracking is the most serious defect in weld since it can lead to premature failure particularly in a dynamically loaded component.
- Crack occurs due to temperature gradients that cause thermal stresses in weld zone, variation in the composition of the weld zone.
- This can be prevented by changing the joint design, by changing the parameters, procedures and the sequence of the welding process.
- It can also be prevented by preheating the component and by avoiding rapid cooling of the weld components.

### 2.21.7 Lamellar Tears

- The work piece when tested in its thickness direction is weaker due to alignment of nonmetallic impurities and inclusions.
- In welding rolled plates, these lamellar tears will appear because of shrinkage of the restrained components during cooling.
- This can be avoided by providing free shrinkage of the members.

### 2.21.8 Weld Spatter

- Due to splashing of the liquid metal on the base metal, rough surfaces are produced known as spatter.
- This is due to high welding current, too low welding speed and arc blow.

**EXAMPLE : 2.25**

Match List-I (Weld defect) with List-II (Causes) and select the correct answer.

| List-I | List-II |
|---|---|
| A. Spatter | 1. Damp electrodes |
| B. Distortion | 2. Arc blow |
| C. Slag inclusion | 3. Improper cleaning in multipass welding |
| D. Porosity | 4. Poor joint selection |

Codes:

| | A | B | C | D |
|---|---|---|---|---|
| (a) | 4 | 1 | 3 | 2 |
| (b) | 3 | 4 | 1 | 2 |
| (c) | 2 | 4 | 3 | 1 |
| (d) | 2 | 3 | 4 | 1 |

**Solution: (c)**

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